

GLOBAL

; up · down · / left/right · **ENTER** select · **DEL** back · Home: 2-col grid, letter jumps, // hop columns · { } page · b screenshot · h help (Help links: g glossary+math · m user guide · s sat history · t tech help · l learn theory · f band plan)

HOME

ENTER opens item; menu scrolls: Satellites, Next Passes, Passes, Track, World Map, Sun/Moon, Space Wx, Weather, Activations, Overhead now, Grid dist/bearing, QRZ Lookup, Location, Update, Settings, Log, Messages, About, Charge/Sleep

SATELLITES

f favorite · v favs-only · n new GP sat · x del manual sat · e EQX table · k OSCARLOCATOR · 3 3D globe · 2 sat-to-sat · o orbital · y sim · t transponders · d 10-day · i illum · **ENTER** passes · right edge: dot = AMSAT heard, square = telemetry, ring = not heard

SAT-TO-SAT (Sats → 2)

Windows when selected sat + a 2nd fav are BOTH above your horizon (start, dur, peak el each). n next sat · r recompute · ` back

LIVE GPS POS (Location → v)

DMS lat/lon (max precision) + decimal, alt, grid, speed, course. Rovers/portable. ` back

3D GLOBE (Sats → 3)

Wireframe Earth, auto-follows selected sat. Graticule, coastline, day/night terminator, QTH, favs, selected sat + footprint + ground-track trail. g DX 2nd location (grid) + footprint · G clear · arrows turn · **ENTER** re-follow · ` back

EQX TABLE (Sats → e)

Equator-crossing UTC + longitude (W-positive) for OSCARLOCATOR, next 3 days. ;/. scroll · d asc/desc node · r recompute · ` back

OSCARLOCATOR (Sats → k)

Live azimuthal-equidistant plot: sub-point, footprint, QTH range ring + full ground track (AOS/LOS). m toggle polar (default, auto N/S, flips at equator) / QTH-centred · ` back

DX DOPPLER TABLE (Mutual → ENTER → d)

RX/TX dial freqs for BOTH stations every 30s across a mutual window. Two lines/step: me (green) + DX (cyan). From the Mutual list, **ENTER** opens a polar detail (me+DX arcs, AOS/LOS/el) then d the table (or d straight from the list). t cycle transponder · m mode: true rule / fixed DL / fixed UL · a anchor dial (me/DX RX/TX) · // in fixed mode step anchored dial to round 1 kHz (else passband 1 kHz) · header shows pb +/- from center · ;/. scroll · ` back

NEXT PASSES (favs)

ENTER track · m world map · t sky-at-a-glance timeline (bars by elev) · r refresh · z deep-sleep until AOS

PASSES (sel)

* = optically visible. ;/. select · d detail · t/ENTER track · n add TX · r recompute · x mutual-DX · v 10-day chart · V visible-pass list · i illum · g grids · w US states · e DXCC

TRACK (sel)

` exits to previous screen & KEEPS radio/rotator tracking (green RAD/ROT/R+R in header); r/o to stop. m TUNE/CAL · d tune mode (FULL/DL/UL/hold) · t next TX · n jump to beacon · c CTCSS · N sat note · k CW both legs (linear) · r radio · o rotator · p polar · a point-here arrow · z big readout · y tilt on/off (ADV) · f Manual · l log QSO · v voice memo (SD) · g grids · w states · e DXCC now · ix2 report Heard (AMSAT) · **ENTER** save cal

BIG READOUT (z from Track)

Big RX/TX + az/el + tune mode (follows Track). Radio+rotator keep tracking · // tune · s/x step/ctr · m/d mode · t TX · n beacon · k CW (linear) · r radio · o rot · y tilt · l log · z/` back

MANUAL (no radio)

Fix one leg, read Doppler freq to tune the other by hand. u toggle HOLD/TUNE leg · // passband (linear) · m CAL · t next TX · z big view · l log · p polar · g grids · w states · e DXCC · `// Track

MANUAL BIG (z from Manual)

HOLD/TUNE legs in big digits · u swap leg · // tune · s/x · m CAL · t TX · z/` back

TRACK · TUNE

// tune -/+ · s step 100/1k/5k · x recenter · tilt to tune if enabled (ADV)

TRACK · CAL

// downlink · ;/. uplink · s step 10/100/1k · x zero

WORKABLE GRIDS / STATES / DXCC

Footprint coverage (per-pass union or live now), count on a cyan line: g 4-char grids · w US states+DC · e DXCC (all 340, hybrid polygons+points). On grids, f prefix filter (EM/EM2/EM21, upper-cased), c clear. ;/. & { } scroll · ` back

POLAR / PASS DETAIL

Pass detail: p polar of this pass. Polar: l log QSO · v voice memo (SD) · p/ENTER/` back

SUN / MOON

Graphic sky-dome (Sun/Moon glyphs by az/el) · g graphic/list · ;/. pick Sun/Moon · o rotor track on/off · s sky sources · t transits · e EME · x stop · ` back

SKY SOURCES (Sun/Moon → s)

Planets (cyan dots) + strong radio sources (orange +): Cas A, Cyg A, galactic centre, Crab, Virgo A, on a sky dome. Antenna-pointing / RF reference. ;/. select · ` back

EME / MOONBOUNCE (Sun/Moon → e)

Self-echo Doppler per band (50/144/432/1296/10368, topocentric) · range + rate · path degradation vs perigee · galactic sky-noise flag · SUN flag <10° · m mutual window · p 30-day plan · o rotor track Moon · ` back

GRID DIST/BEARING (menu)

Enter Maidenhead grid → great-circle distance + beam heading (short/long path, km/mi). g grid · q QRZ→grid lookup (seeds calc) · o point rotor at bearing · ` back

SPACE WX (menu)

Solar 10.7cm flux + planetary Kp + A index + aurora likelihood (from Kp), labelled & colour-coded, with HF/sat operating outlook & data age · **p** HF/6m propagation · **r** refresh (WiFi) · **`** back

HF/6m PROPAGATION (Space Wx → p)

Turns solar flux + Kp into band guidance: HF conditions (10/15/20m open/marg/shut), geomagnetic effect, auroral-VHF likelihood (6m/2m, beam N), D-layer absorption. Rule-of-thumb · **r** refresh · **`** back

WEATHER (menu)

Current conditions + multi-day forecast for your site (Open-Meteo). Refreshes on entry (WiFi) & with Update. Units in Settings · **r** refresh · cached offline · **`** back

QRZ LOOKUP (menu)

Callsign lookup via QRZ.com XML (needs QRZ XML subscription + user/pass in Settings → Network). **ENTER** type call → name/addr/grid/class. WiFi req'd · **`** back

BAND PLAN (Help → f)

Worldwide amateur band reference LF→light: HF with ITU R1/R2/R3 splits, VHF/UHF/microwave EME+calling freqs, satellite subbands, IARU designators (H/A/V/U/L/S/C/X/K), sat modes incl QO-100. **;/**, scroll · **`** back

ACTIVATIONS (menu)

Upcoming sat activations on hams.at (roves, grid/special ops). List: date, call, sat, grid (* = your own entry). **;/**, move · **ENTER** detail · **n** add your own sked (offline OK), **e** edit a * entry · **r** refresh · **`** back. Detail: UTC/mode/freq + **footprint note** (checks co-visibility with the activator +/-30 min of listed time). **;/**, scroll the full comment, a SKED reminder (T-60/30/10 beeps+flash), **w** mutual-window screen if a footprint exists. Mutual window: small polar plot (me + DX arcs), Date/AOS/L/O/S/dur + peak el each; **DX** Doppler pre-set to the transponder & fixed DU/UL parsed from the freq/comment (default table if none). Cached to card — last list shows offline; WiFi to refresh

OVERHEAD NOW (menu)

Snapshot of every catalog sat above the horizon right now, sorted by elevation, with az + rise compass dir (high=green, low=yellow) + count up/scanned. **r** rescan (this instant) · **;/**, scroll · **`** back

TRANSPONDER DB (Sats → t)

Sat's transponder/beacon entries, two lines each: **D**=downlink+mode line, **U**=uplink+tone/inv line. **;/**, select (* = manual) · **x** del manual (2x) · **`** back

ORBITAL ANALYSIS

;/ 10 pages: Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position / Phys (velocity + launch date/age) · **r** recompute · Doppler **f** sets beacon freq

SIMULATION

;/ step time · **;/**, step size · **m** world-map view (sub-point + footprint) · **x** reset to now · **`** back

LOCATION

e/o/a lat/lon/alt · **g** grid · **p** GPS on/off · **s** GPS source · **e** set clock · **v** live position (DMS) · **ENTER** GPS sky plot

GPS SKY PLOT

Live GNSS by az/el, coloured by signal (green=strong, grey=weak) · **`** back

WORLD MAP

All footprints + night-side shading · **f** highlight favorite · **c** recenter on QTH/0° · yellow=sunlit cyan=eclipse · **`** back

SCHEDULES

10-day: **;/**, scroll +/−1 day (fills full days), **r** recompute · illum: **;/** scroll +/−60 days · Mutual: **;/**, scroll, **ENTER** polar detail, **d** Doppler

LOG

Menu (scrolls): **ENTER** new QSO / browse / export ADIF / LoTW upload / Cloudlog upload / voice memos / notes / awards · **Awards**: QSO/grid/state/DXCC totals (states+DXCC derived from grid), **g/s/d** scrollable worked lists, **ENTER** per-sat · List: **;/**

scroll, **ENTER** edit · Entry: **;/**, field, **ENTER** edit, **s** save, **x**2 delete · editing a QSO re-arms its upload; extra **LoTW/Cloudlog** rows (ENTER toggles) override that

SIGN & UPLOAD TO LoTW (Log → Sign & upload)

Uploads sat QSOs direct to ARRL LoTW over WiFi. Needs SD + your LoTW key (lotw_key.pem/lotw_cert.pem in /CardSat/ — make them from your TQSL .p12 with tools/lotw_cert_converter.html in a browser; in Settings pick DXCC → primary (state/province/...) → secondary (county/city) + zones + IOTA, all gated pickers w/ type-to-filter) — see manual §8. **u** upload · **a** re-send all · **`** back

UPLOAD TO CLOUDLOG (Log → Upload to Cloudlog)

Uploads sat QSOs to a self-hosted Cloudlog/Wavelog over WiFi (an alternative to LoTW — Cloudlog can forward to LoTW). Set **URL** + read-write **API key** + **station ID** in Settings. **u** upload · **a** re-send all · **`** back

VOICE MEMOS (Log → Voice Memos)

Browse newest-first (date/time/sat/len). **ENTER** play · **n** new · **d** del · **r** refresh · record via **v** on Track. SD req'd

NOTES (Log → Notes)

Free-form text notes (.txt in /CardSat/notes/, newest-first w/ date/time). Browser: **ENTER** open · **n** new · **d**+ENTER del. Editor (commands use **Fn** so **;/** type): type freely, **ENTER**=newline, **DEL**=backspace · **Fn+;/Fn+;** cursor L/R · **Fn+;/Fn+;** up/down · **Fn+s** save · **`** exit (auto-saves)

LORA MESSAGES (Home → Messages)

CardSat-to-CardSat broadcast chat (Cap LoRa). Same freq/SF/BW = same group. Set **region** (US 33cm / EU 70cm / JP 430) in Settings for a legal default freq. **n** write/send · **;/**, scroll-select (newest on bottom) · **o** station roster · **r** retry · **m** LoRa RX/hex monitor · **`** back. **Actionable msgs** (plain text, interop): a msg with @lat,lon + **#SAT** / **ISAT date time** → **ENTER** opens bearing compass (dist/brg/grid) / sat detail / pre-filled sked; **#SAT** carries NORAD (**#name/norad**) so it resolves across differing names. Send for the current sat & your location: **p** position (also a presence ping) · **s** satellite · **k** sked (date→time). **Roster (o)**: who is heard, with grid + dist/brg + signal; **ENTER**=compass, **p**=ping. **Opt-in Auto position reply** setting answers others' positions (loop-guarded). **Rx always on**: badge on any screen + banner (opt-in beep) — **Msg notify** in Settings. Needs RadioLib build.

LORA RX / HEX MONITOR (Messages → m)

Receive/inspect any LoRa signal (not just sats). Config: set freq (type in **MHz**), SF, BW, CR, sync, preamble, CRC — saved across reboots. **ENTER** starts RX. Monitor: live hexdump+ASCII, RSSI/SNR, **p** pause (read a frame on a busy channel) · **;/**, scroll · **s/b/c/f** tune SF/BW/CR/step · **;/f** freq · **`** esc. Rx-only. Untested HW

UPDATE

k/ENTER GP+clock+AMSAT+space-wx+weather · **f** fast (GP+AMSAT+favs' TX) · **a** cache all TX (auto-reboots) · **w** WiFi only

SETTINGS

;/ change · **ENTER** edit/toggle · **s** scan WiFi · opt. 2nd WiFi (field fallback) · **reset** = ERASE

GP SOURCE

AMSAT / **CelesTrak** JSON-PP category / **Custom URL** · **;/**, move · **ENTER** select

ROTATOR (manual)

;/ az · **;/**, el · **s** step · **x** stop · **`** back

NETWORK SERVERS

rigctd PC drives rig (VFOA=DL/B=UL) · **rotctd** PC drives GS-232 · **rigctl** drives remote rig · **Web** opt-in mobile page (no auth, trusted LAN): live sky plot, Doppler readout, tap-to-copy freqs, visible-pass list + AOS alerts, radio/rotator control

EDIT

type · **DEL** backspace · **ENTER** ok · **`** cancel

ABOUT

Build/version, IP, free heap ads and diagnostics (read-only). **r** Station readiness checklist · **t Tools** (20): calculators, DXCC/CQ/ITU lookups, RF/antenna workbench, link budget, RF exposure, orbit lifetime · **I** License & credits (disclaimers, data sources, support AMSAT) · **z Games menu**: six mini-games (**;/**, pick, **ENTER** launch) — Zap the Sats, Doppler Lock, Catch the Pass, Rotor Runner (tiltkeys), Morse Meteors (t dot u dash), Grid Chase.